

fit@hcmus

Software Testing

CSC13003

Software Testing Introduction

Content

- What is Software Testing?
- Why must we test?
- What does a Software Tester do?
- Potentials of Software Testing Career?
- Tester is a “low-level” job?
- How is a good tester?

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Software Testing

Testing is the process consisting of all lifecycle activities, both **static and dynamic**, concerned with planning, preparation and evaluation of software products and related work products to determine that they **satisfy specified requirements**, to demonstrate that they are **fit for purpose** and to **detect defects**.

Source: Glossary of Software Testing Terms

Static Testing vs. Dynamic Testing

- Static Testing

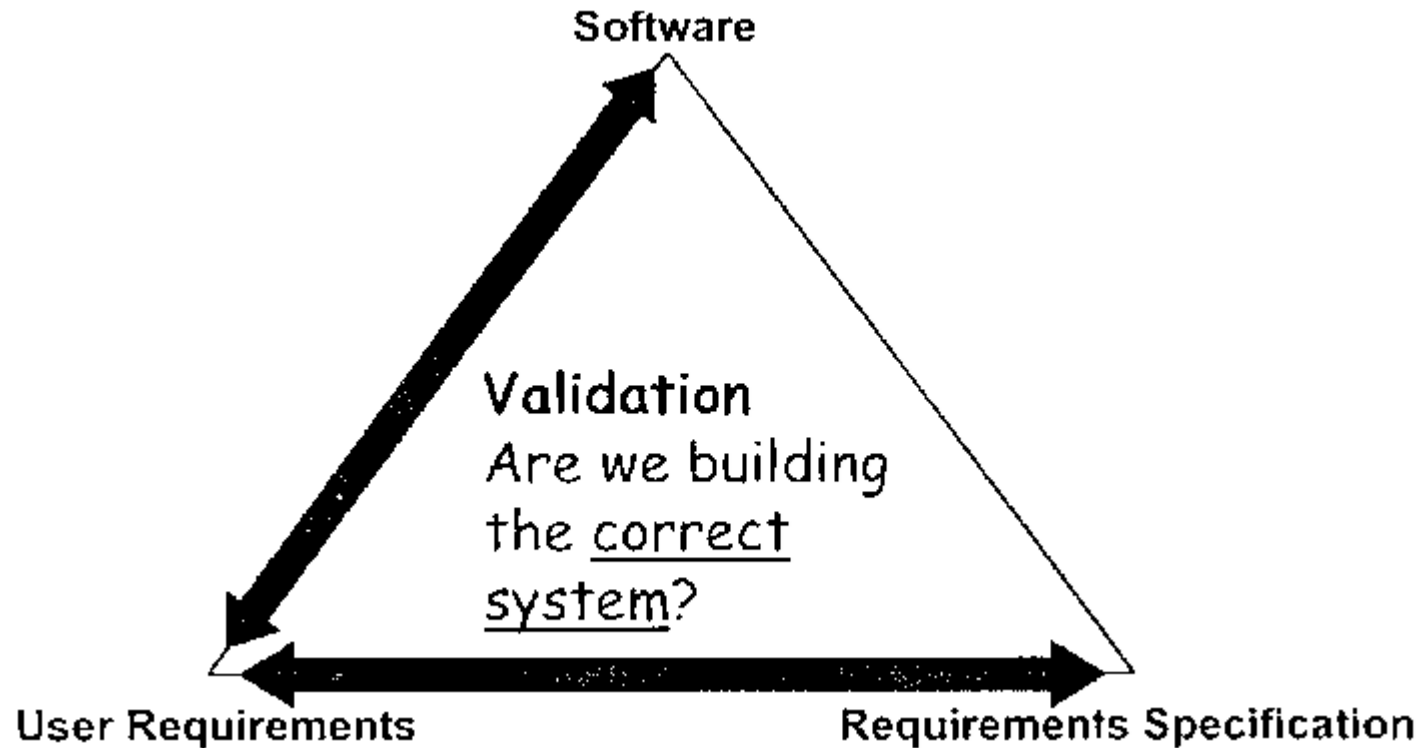
- Examine the code
- NOT Execute it
- Code analysis, Code inspection, Code review...

- Dynamic Testing

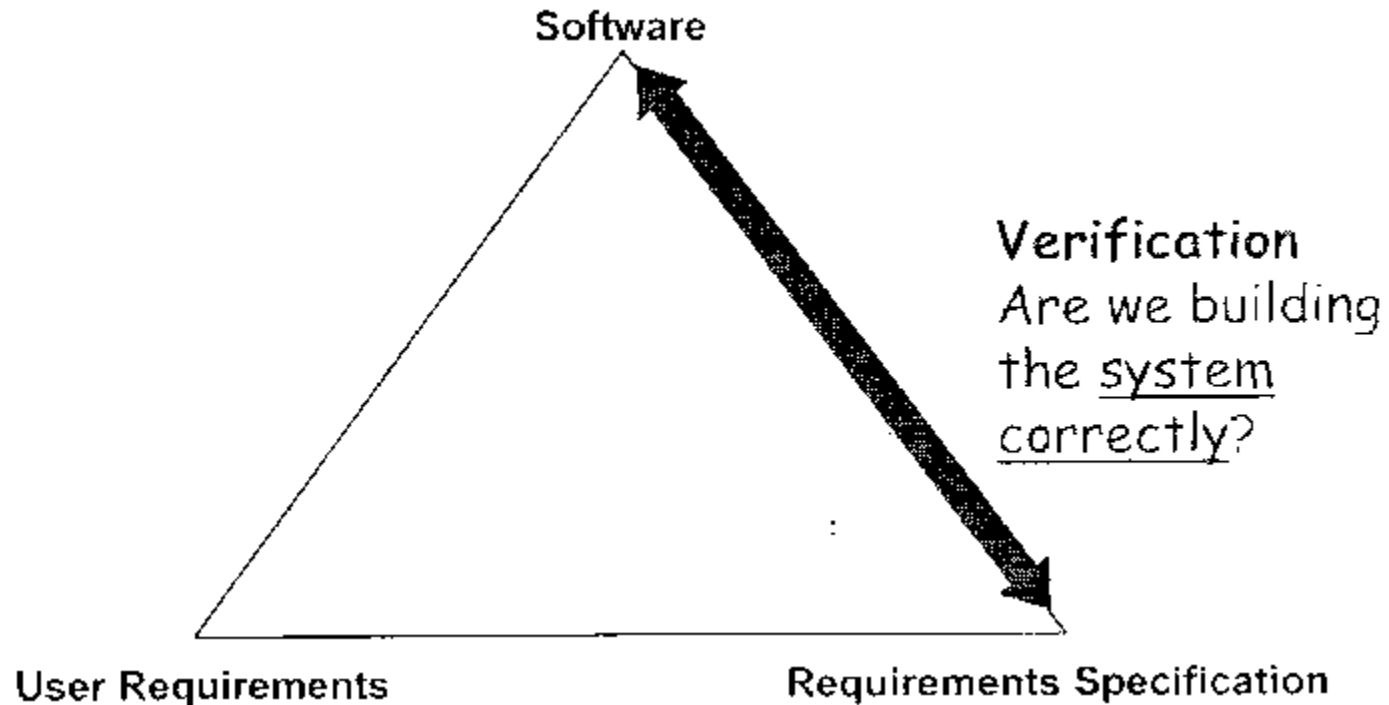
- Execute the code
- Without, necessarily, examine it



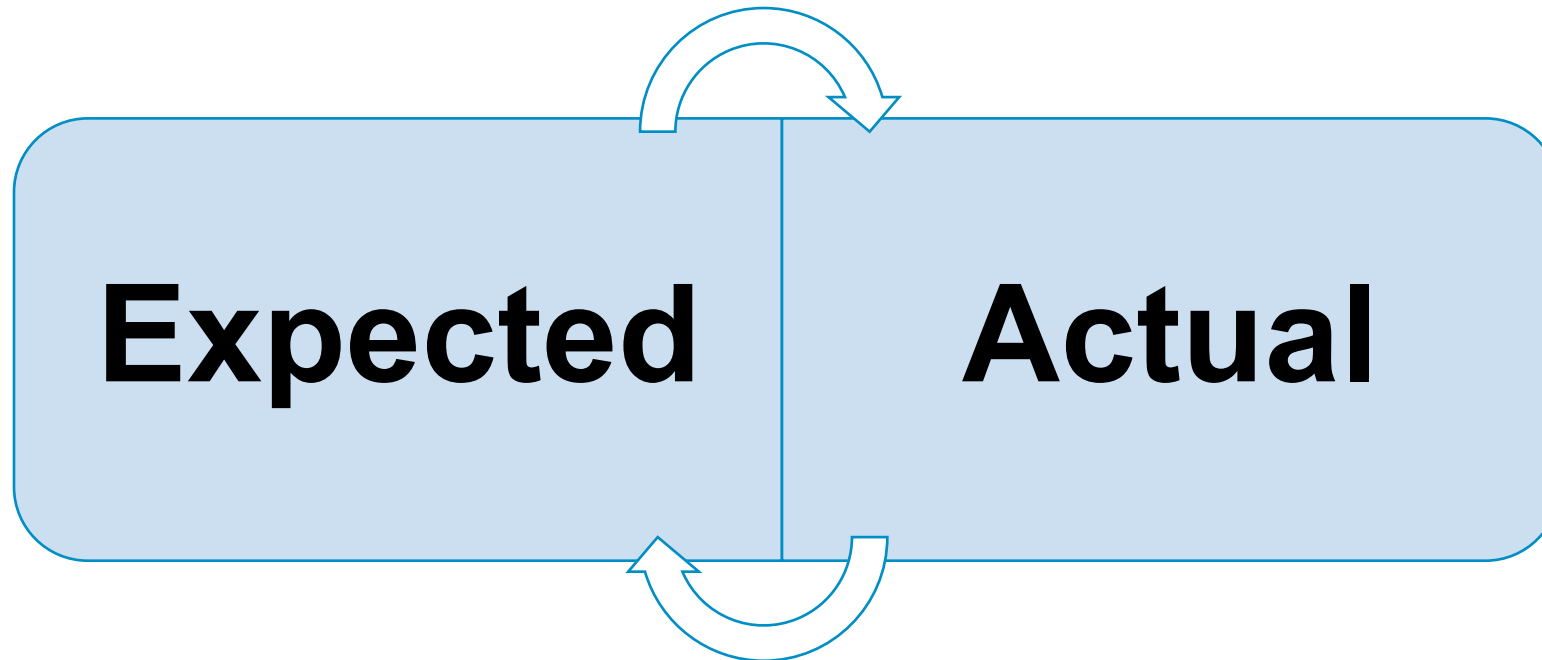
Validation Testing vs. Verification Testing



Validation Testing vs. Verification Testing



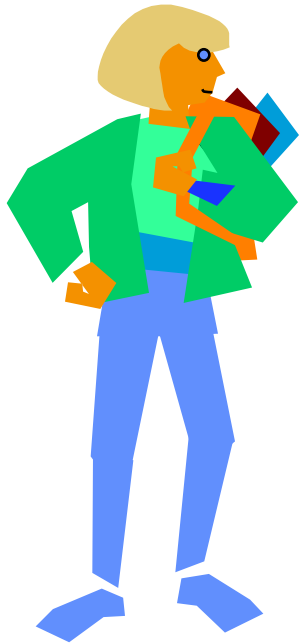
Test Oracle



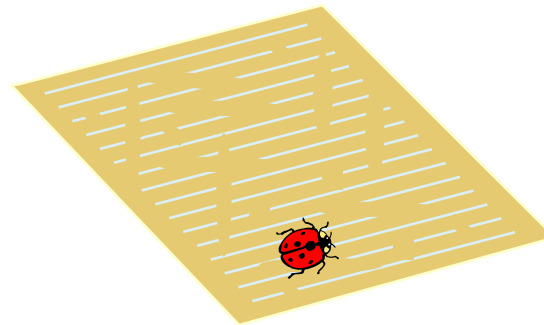
- **Test Oracle:** a source of information about whether the output of a program (or function or method) is correct or not

Error – Fault – Failure

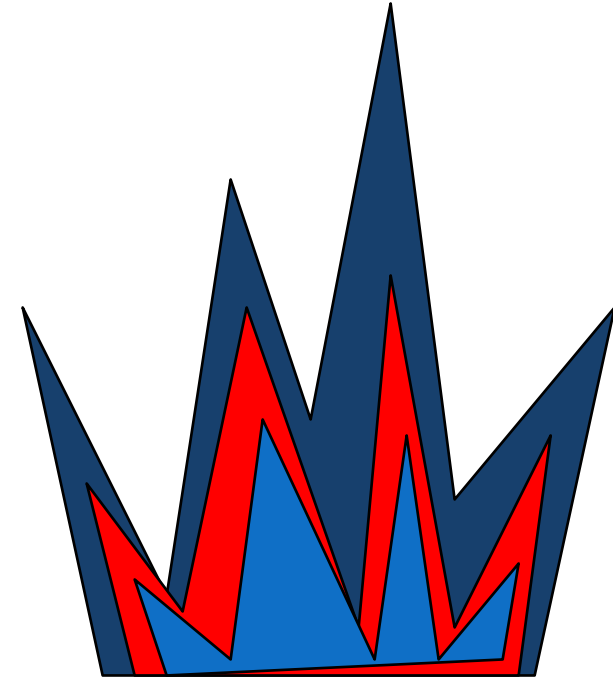
A person makes
an **error** ...



... that creates a
fault in the
software ...



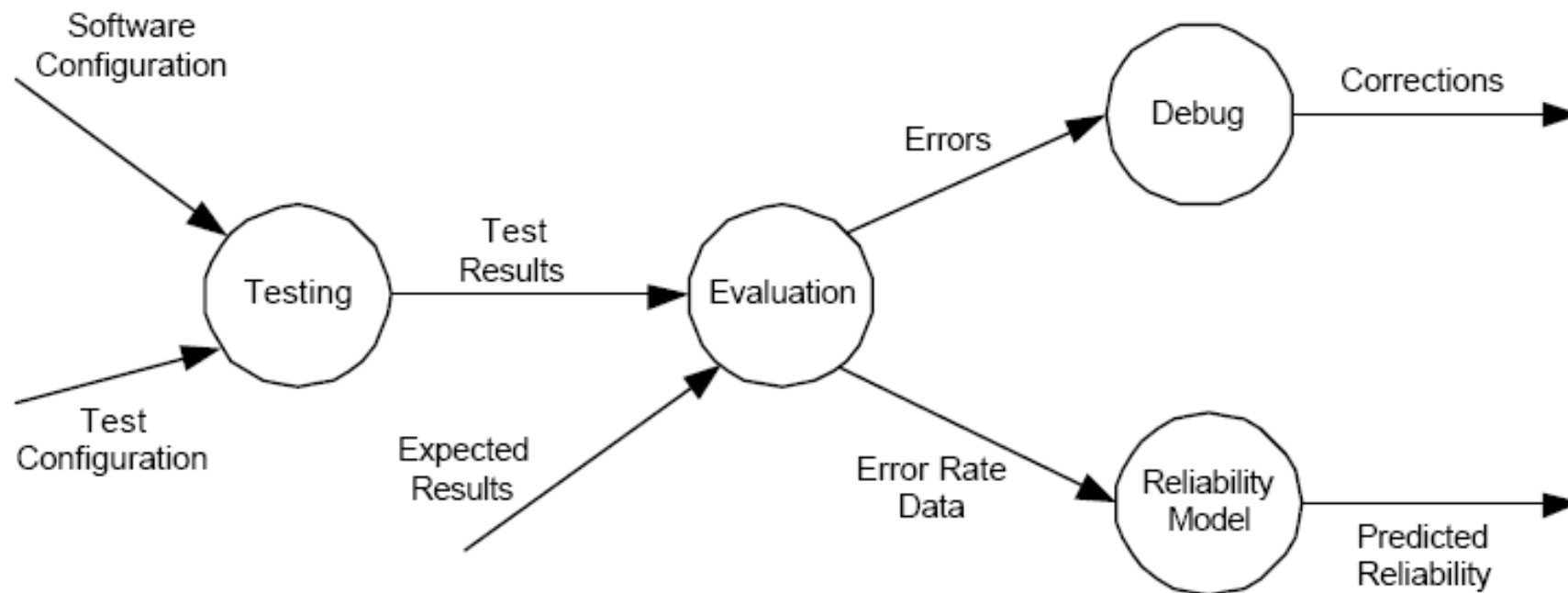
Fault / Defect / Bug



... that can cause
a **failure**
in operation

Testing vs. Debug

- Testing: finding input that cause the program to fail
- Debug: the process of finding a fault given by a failure



Test Bed

- The test execution environment configured for testing
 - Hardware
 - Software
 - Operating system
 - Network configuration
 - The product/application under test (AUT)
 - Other system/application
 - ...

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How the customer explained it



How the Project Leader understood it



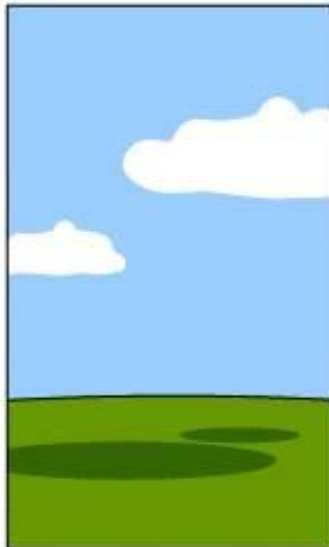
How the Analyst designed it



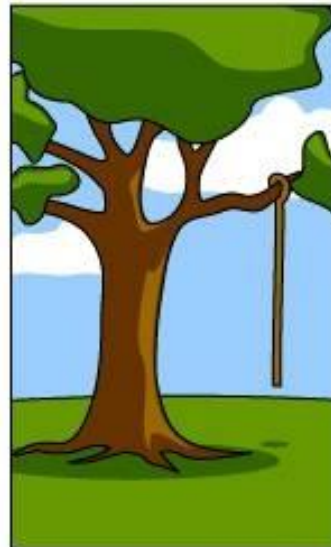
How the Programmer wrote it



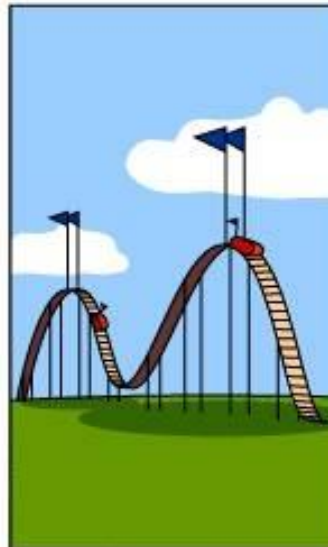
How the Business Consultant described it



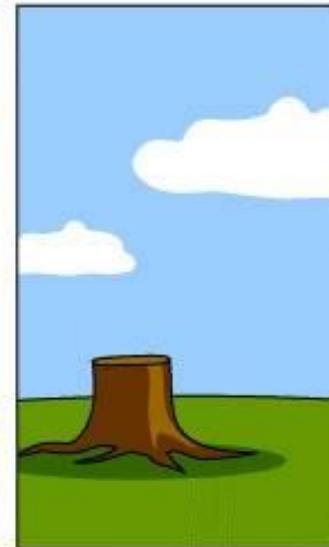
How the project was documented



What operations installed



How the customer was billed



How it was supported



What the customer really needed

Why must we test?

“A clever person solves a problem. A wise person avoids it”

Albert Einstein

Software bugs could be expensive and even dangerous

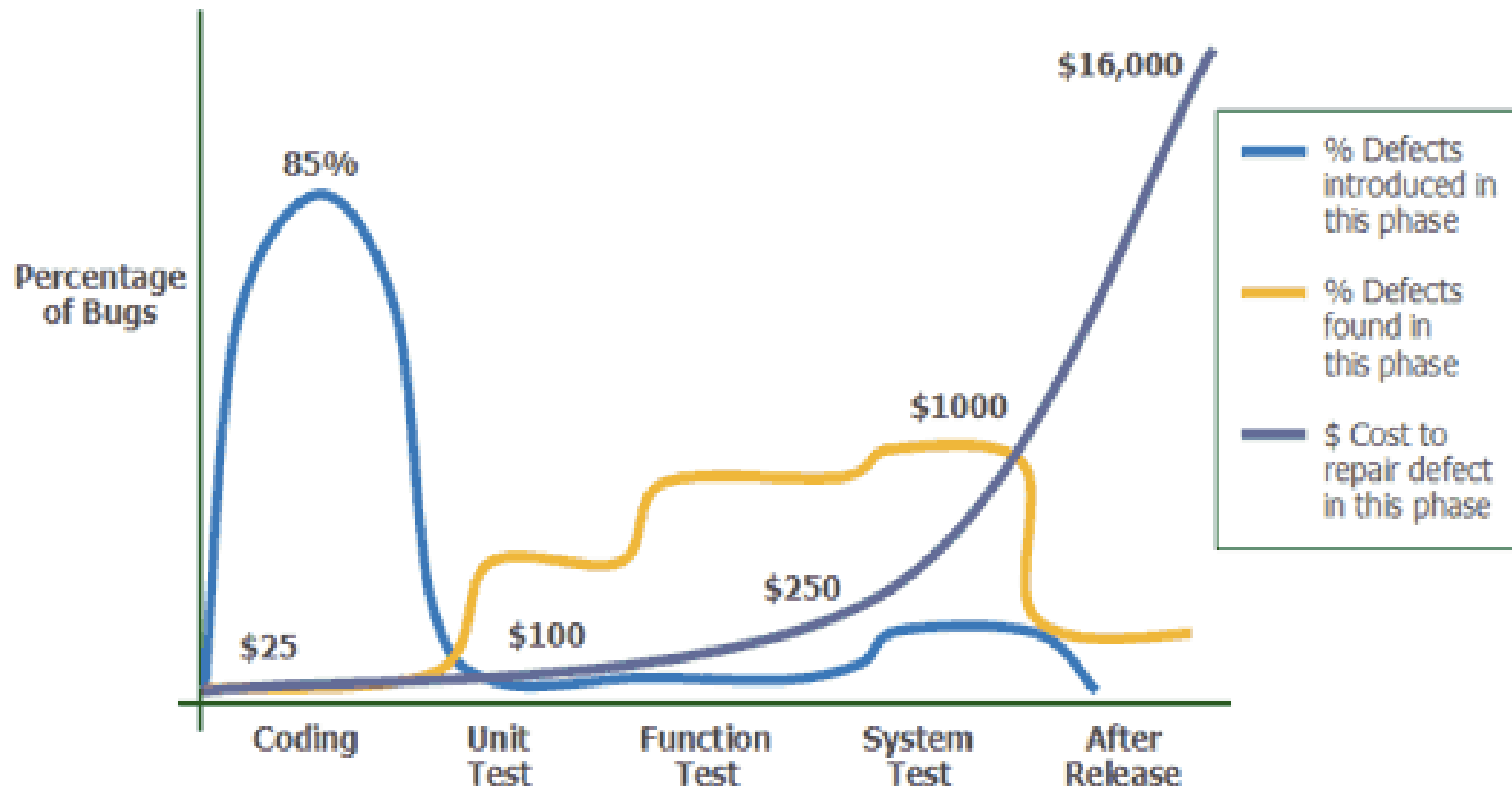


Korean Air Flight 801 crashed, 1997
254 people dead & injured



Ariane 5 rocket exploded, 1996
\$500 million lost

Cost of fixing defects



Objective of Testing

- Detect Faults
- Establish confidence in the software
- Evaluate properties of the software
 - Reliability
 - Performance
 - Memory usage
 - Security
 - Usability
 - ...

Objectives of Testing

- Should NOT be to verify that the program works correctly
 - If you can't test the program completely, you can't verify that it works correctly
 - “Testing can show the presence of bugs, but not the absence” - Dijkstra

If you set your mind to show that a program works correctly, you'll be more likely to miss problems than if you want and expect the program to fail.

Objectives of Testing

The Objective of Testing a Program is to Find Problems

Finding problems is the core of your work. You should want to find as many problems as possible. The more serious testers find, the better tester is.

A test that reveals a problem is a success. A test that did not reveal a problem is (often) a waste of time.

Objectives of Testing

The Purpose of Finding Problems is to Get Them Fixed

The point of the exercise is quality improvement!

- The best tester is not the one who finds the most bugs or who embarrasses the most programmers.
- The best tester is the one who gets the most bugs fixed.

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Comparison between QA, QC and Testing



- Subset of SDLC
- Process oriented
- Ensure that processes and procedures are in place to achieve quality
- Focus on process to achieve required quality

- Subset of QA
- Product oriented
- Activities to ensure the product quality
- Focus on product to check for the required quality

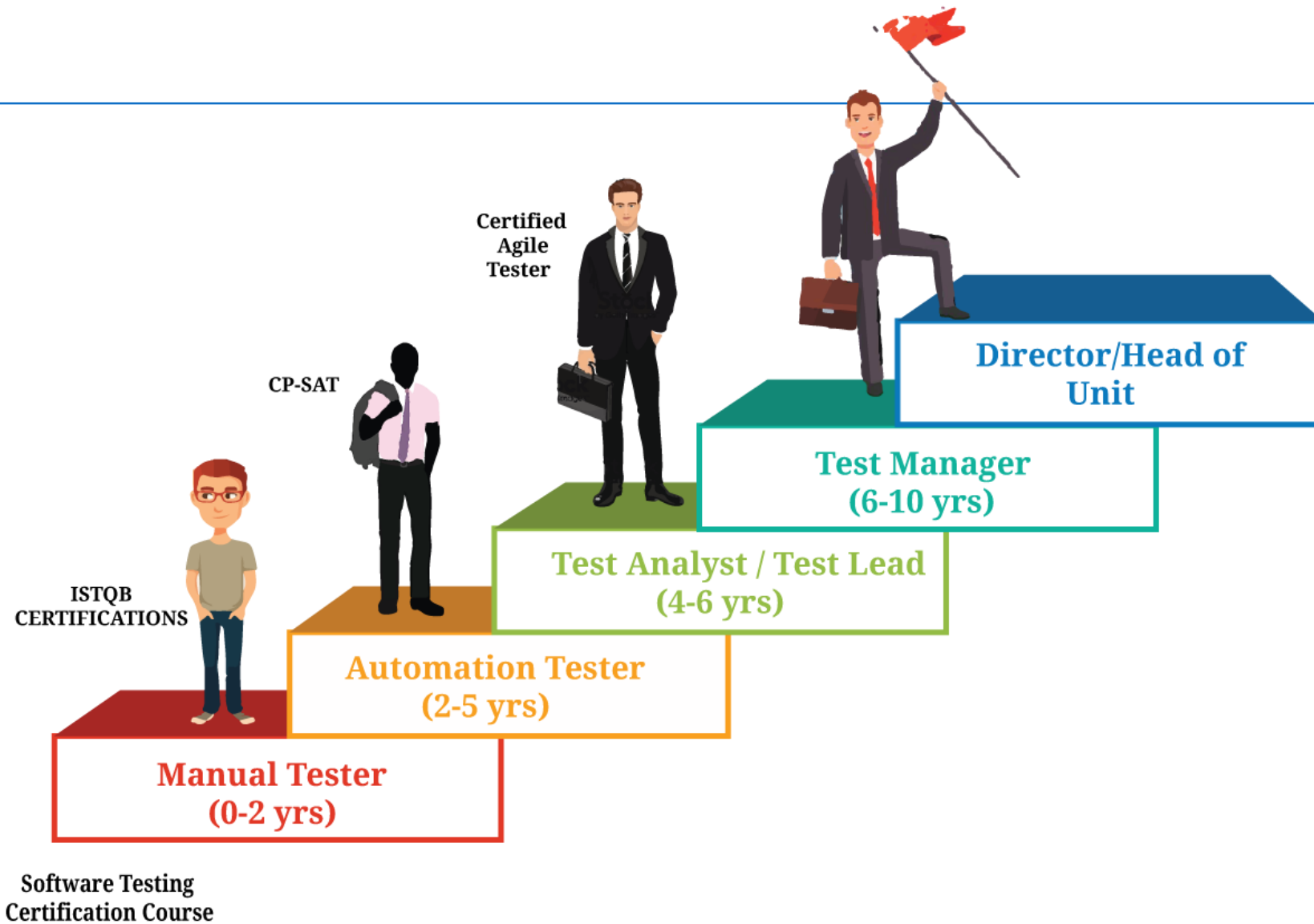
- Subset of QC
- Product oriented
- Validate the product against specifications
- Focus on actual testing of the product

QC vs. QA

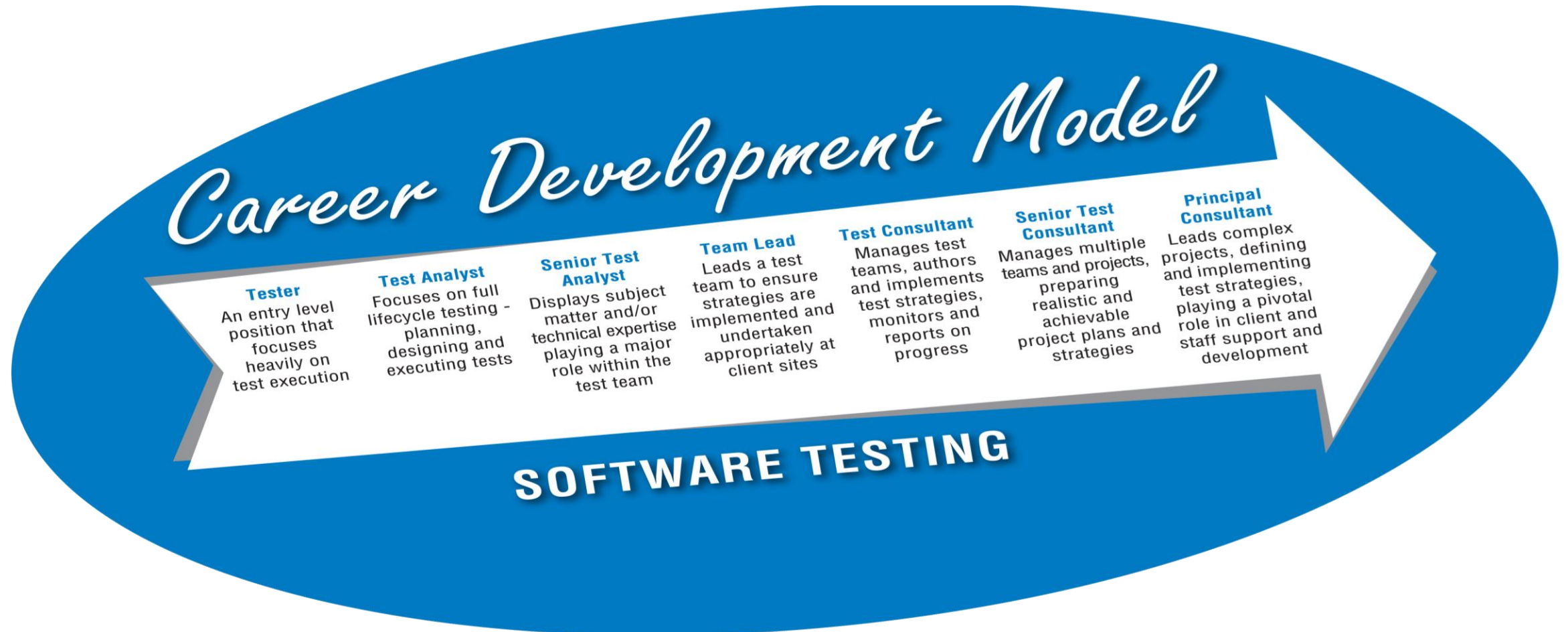


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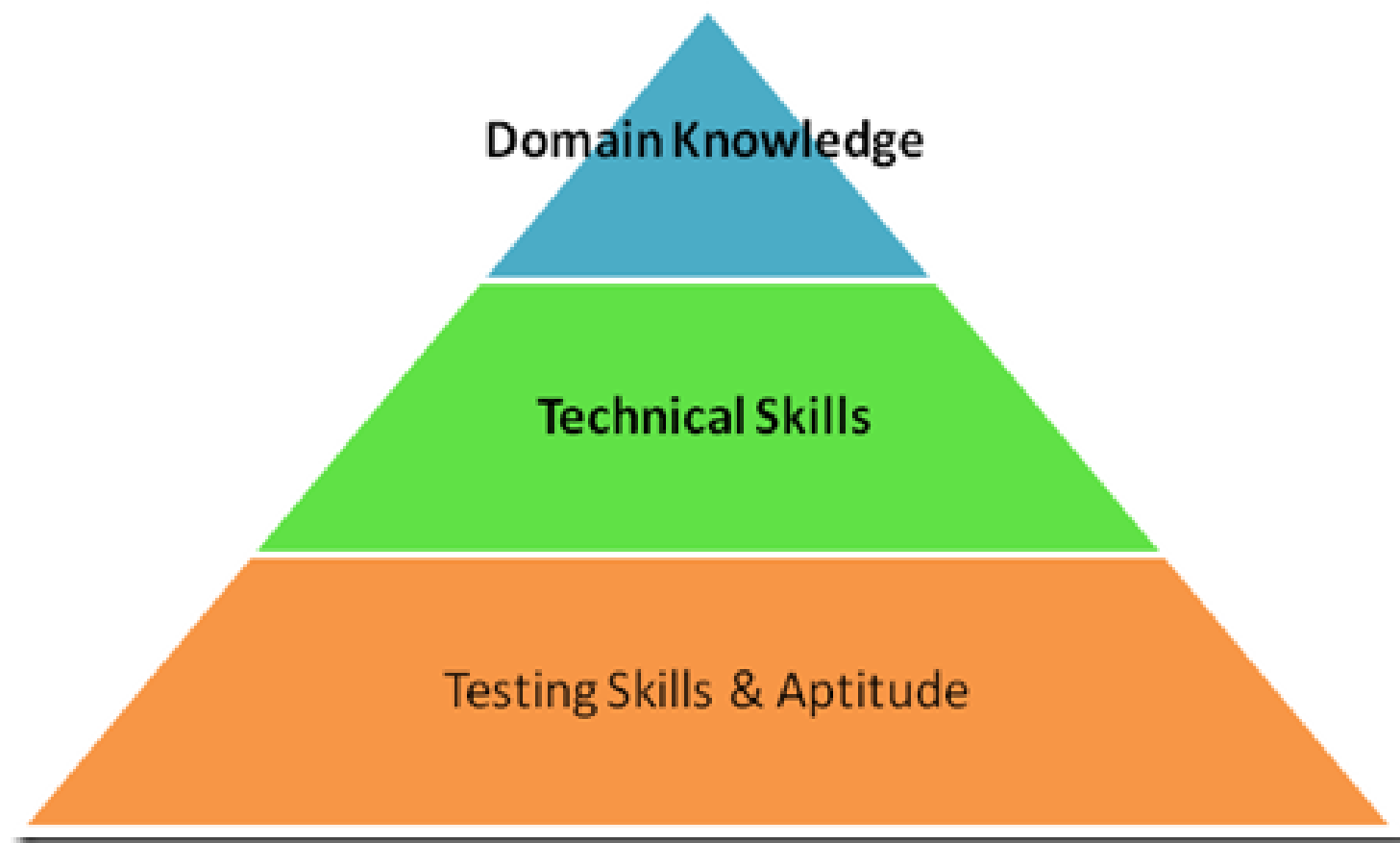


Software Testing Career Path



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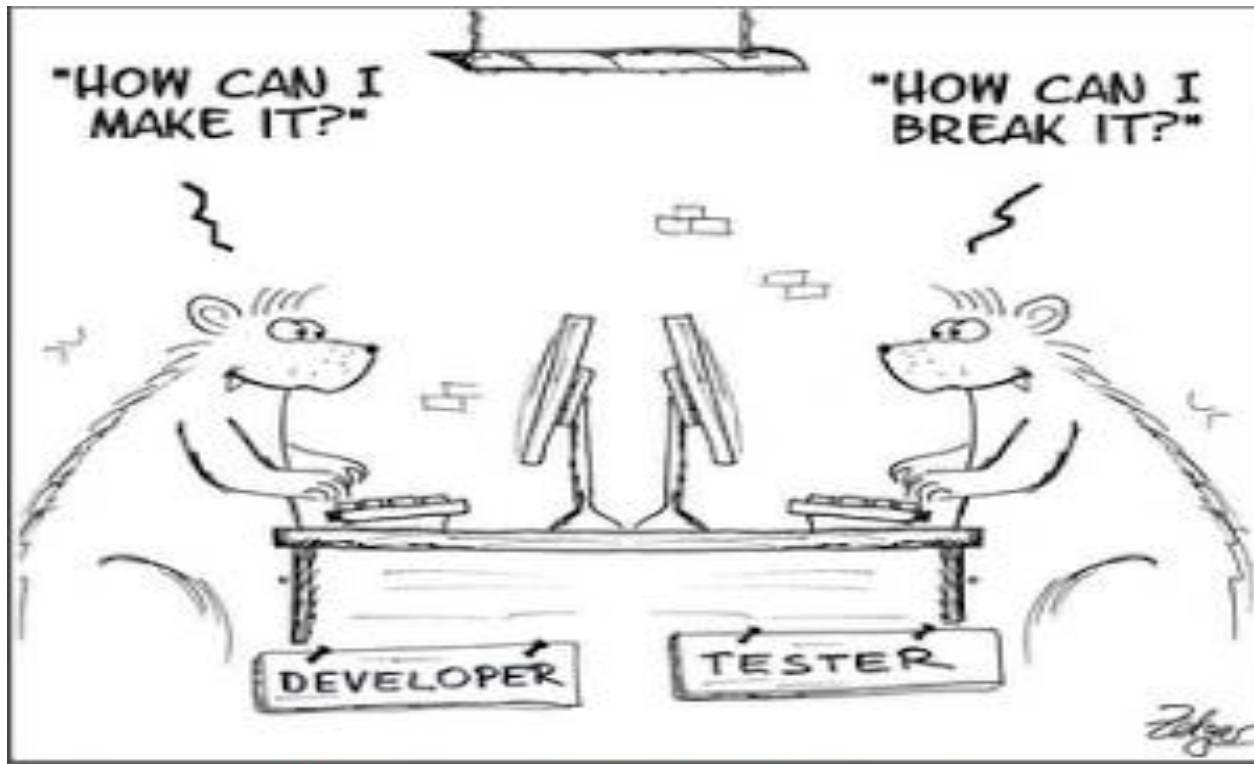
Domain Knowledge



Good Software Tester



Tester vs. Developer



They are not so much different,
but they have different path for the same goal,
to improve quality!!

